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**A Machine Learning Approach for
Customer Risk Segmentation in Digital
Lending**

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Declaration

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Abstract

In the evolving landscape of digital lending, the accuracy of credit risk scoring is important for the successful growth of the digital lender and increase in financial inclusion to borrowers to access credit. The study endeavors to address this critical need by leveraging machine learning tools to enhance the precision and granularity of the customer credit risk segmentation. Furthermore, the study conducts a comparative analysis between top-performing machine learning models and conventional statistical methods, to offer valuable insights into the potential improvements achievable through machine learning methodologies. The study will also explore model interpretability techniques, such as partial dependence plots and Shapley values, to understand the intricate factors influencing credit risk assessment outcomes. The practical application of these techniques aims to shed light on the rationale behind risk assessments, enhancing transparency and providing actionable insights for decision-makers in the lending domain. Through these endeavors, the research aims to contribute to a more robust and data-informed approach to credit risk assessment, benefiting both lenders and borrowers in the digital lending ecosystem.

Keywords: Digital Lending, Machine Learning, Credit Risk Scoring, Customer Segmentation, Machine Learning Interpretability.

Chapter 1

Introduction

1.1 Background

The recent rapid evolution of Financial Technology (Fintech) has played a key role in reshaping how financial products and services are generated, delivered, and utilized due to its potential to transform the entire financial system. This has been largely attributed to the massive surge in data availability which has created innovative opportunities for its utilization. Big data has been acquired and put to widespread use in artificial intelligence (AI) and machine learning (ML), with applications spanning digital lending, credit risk scoring, fraud detection, and various other domains (Allen, Gu, and Jagtiani, 2020).

1.1.1 Digital Lending

In recent years, the digital lending sector has experienced remarkable expansion, propelled by technological advancements, shifting consumer preferences, and a growing appetite for easily accessible financial services (*Digital Lending Market Size, Trends , Growth — Forecast - 2032* 2023). Consumer demand for digital financial products is rising as the innovations in Fintech continue to progress. Globally, the unbanked have an annual income opportunity of about \$380 billion leading to a significant demand for short-term, unsecured lending from consumers and small enterprises around the world ((Davis,

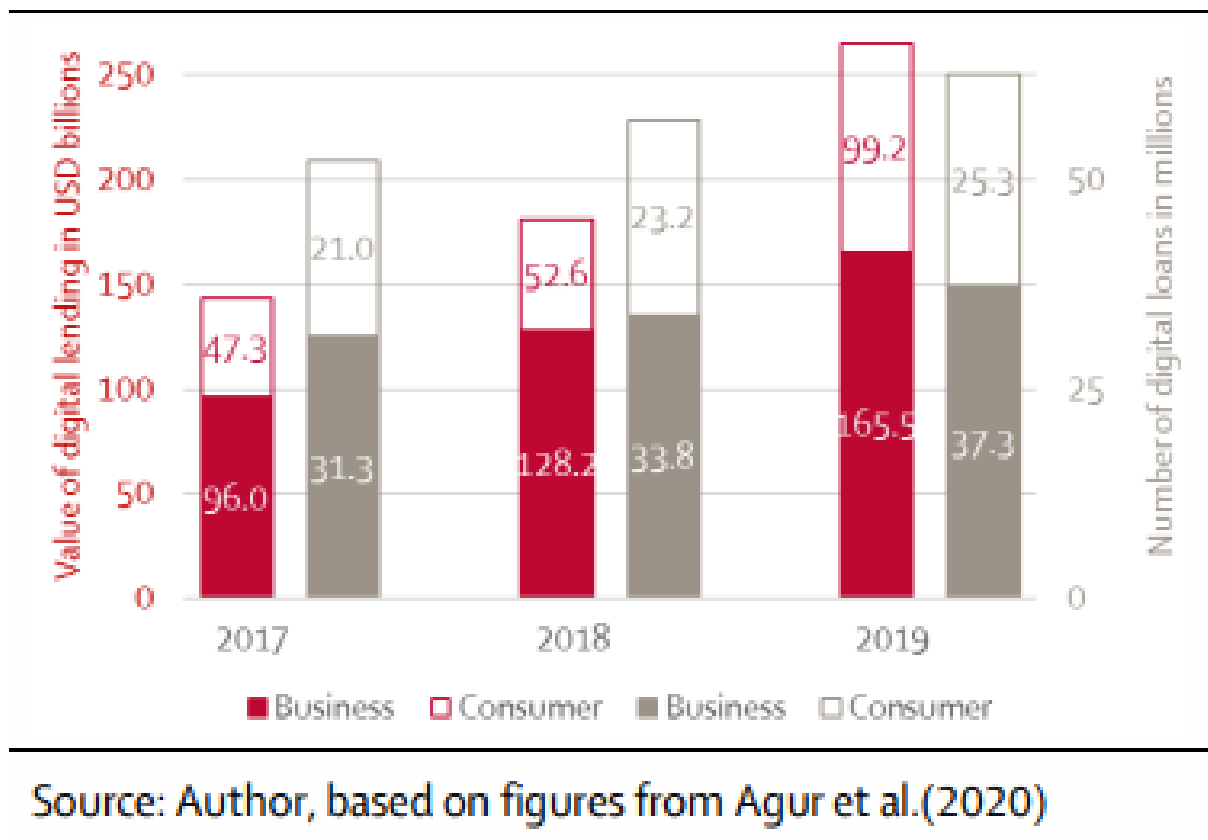
2019).

Since the inception of M-Shwari in 2012, the digital lending industry has witnessed significant expansion in both the number of digital lenders and the volume of loans disbursed. Progress in credit scoring, minimal regulatory hindrances, and the widespread adoption of mobile phones and mobile money have facilitated the robust growth of digital lending. This growth provides borrowers with a swift and user-friendly alternative for accessing credit (Paul, 2019). During the COVID-19 pandemic, digital lending experienced substantial growth as people and enterprises sought the convenience and contactless means of accessing financial resources (Data, n.d.).

Since 2005, a number of Fintech companies have entered the digital lending industry with prominent digital lending platforms including Lending Club, Zopa, Prosper, and Peerform. In addition to these established names, Tala stands out as a unique lending platform that specializes in granting micro-loans, ranging from \$10 to \$500, to borrowers in developing countries who lack a traditional credit history. Remarkably, Tala boasts an impressive repayment rate exceeding 90% Hindistan et al., 2019.

In Kenya, a leader in digital finance, the number of digital lending has already surpassed that of traditional loans with an astounding 86% of loans obtained between 2016 and 2018 (*Making Digital Credit Truly Responsible: Insights from analysis of digital credit in Kenya* 2019). Globally, fintech companies saw a 57% increase in their digital loan portfolios worldwide between 2017 and 2019, as illustrated below (with the numbers being conservative estimates due to incomplete data) (Christoph, 2021). As much as there's an increase in the usage of digital lending, traditional lending still maintains a significantly larger portfolio size due to the typically small and short-term nature of digital credits (Christoph, 2021).

In 2018, the overall approximate value of digital loans provided to borrowers amounted to KShs 116.8 billion (\$1 billion), averaging KShs 37,464 (\$324) per digital borrower annually (Paul, 2019). In 2022, the global digital lending market had a valuation of \$12.6 Billion, and it is anticipated to attain a figure of \$71.8 Billion by 2032. This rep-



Growth in Digital Lending (Fintechs only (Christoph, 2021)

resents a notable growth trajectory, with a Compounded Annual Growth Rate (CAGR) of 19.4% projected from 2023 to 2032 (*Digital Lending Market Size, Trends , Growth — Forecast - 2032* 2023).

The growing need for digital lending services has opened doors for platform providers to seek funds to address this demand and broaden their lending operations. Progress in data analytics and risk evaluation methods linked to digital lending platforms bolster investor trust, drawing in additional investments to bolster the industry’s expansion . There are substantial investments directed toward emerging technologies like machine learning (ML) that will empower lenders to automate loan underwriting procedures, scrutinize extensive datasets for risk evaluation, and tailor loan options to match the unique profiles of individual borrowers (Data, n.d.).

Digital lending practices have the potential to compromise the integrity and stability of the wider financial systems. Evidence from various countries show that nonperforming loans (outstanding loan balances not paid for more than 90 days) were increasing even

before the COVID-19 crisis, and this growth is connected to the surge in digital lending. Loan defaults, for instance, were associated with negligent digital lending practices in Kenya and Tanzania. In Kenya, defaults accounted for 16% of all digital credits between 2016 and 2018, compared to default rates of 11.7% for total private credit and nearly half (49%) of the borrowers with non-performing loans still owe less than \$10 of their outstanding sums (Christoph, 2021).

To level the playing field for digital lenders and to maintain the integrity and stability of the credit market, regulators must enact new laws or amend existing ones to require particular licenses and rules for all digital financial service providers. Without regard to their business models, digital lenders must conduct proper creditworthiness evaluations (Christoph, 2021).

1.1.2 Credit risk scoring

Popularly known credit risk scoring system, FICO (Fair Isaac Corporation) scores that was developed in the 1950s held a key role in making critical credit-related decisions, particularly in the context of credit card applications (Hindistan et al., 2019). Earlier research studies, including the investigations by Mester, Nakamura, and Renault, 2007 have underscored the paramount importance of factors such as consumers' credit histories and various financial and accounting data when assessing credit risk for individuals by lending institutions.

Unfortunately, the heavy reliance on traditional credit risk scores has created barriers for certain consumers, either by hindering their access to affordable credit or dissuading them from seeking credit altogether. In the 2019 Survey of Consumer Finances (SCF), (*Federal Reserve Board - Survey of Consumer Finances (SCF)* 2019), reveals that 9.3% of individuals aged 18 and above encountered challenges in their credit applications. Nearly half of these individuals, approximately 45.3%, attribute these credit-related difficulties directly to their credit risk scores. An additional 7.4% of consumers opted not to apply for credit altogether. This is largely due to their limited or nonexistent traditional credit

risk scores as they are new to the credit market or do not bank accounts, and have shorter payment histories Toh, 2023.

In recent years, there has been a notable transformation in credit risk assessment specifically among digital lenders who have explored and initiated efforts aimed at innovative credit-scoring models. These models are designed to leverage the wealth of alternative data sources available today, as well as to explore the application of advanced machine learning techniques. This transformative shift marks a significant departure from traditional approaches and underscores a dynamic evolution in the field of credit assessment (Toh, 2023). These alternative data may include data on bill payments, mobile data transactions (for example, mobile money transactions, mobile phone usage in data, calls, and texts), and income or assets (for example, employment history and property ownership).

Research by Di Maggio, Ratnadiwakara, and Carmichael, 2022 indicates that credit-scoring models incorporating alternative data, whether used independently or in conjunction with conventional credit bureau information, have the capacity not only to assess a broader spectrum of consumers but also to demonstrate equal or superior predictive performance in assessing the likelihood of consumer defaults compared to traditional credit scores. As much as the use of alternative data has its benefits, it has limited adoption by mainstream lenders mainly due to a lack of motivation to use alternative data as they may not be currently lending to these consumers and additionally, they may not have access to this data (Toh, 2023).

1.1.3 Machine Learning

Machine learning techniques in credit scoring models are used to classify consumers into their various credit scores. Additionally, by analyzing their behaviors, consumers are clustered into similar groupings based on their patterns. The alternative data mentioned above on mobile phone data is particularly applicable in customer segmentation where the behavior of a customer is grouped based on the Recency, Frequency, and

Monetary model RFM. These values of recency, frequency, and monetary are combined to form credit risk scores (Maryani and Riana, 2017). The RFM technique relies on three fundamental customer characteristics: the recency of their purchases, how frequently they make purchases, and the monetary value of their purchases. The primary aim of RFM is to forecast future consumer actions, facilitating improved segmentation decisions (Maryani, Riana, et al., 2018).

The inherent complexity of machine learning models often results in a "black-box" scenario, where decisions are made without clear visibility into the underlying factors driving those decisions. This lack of transparency poses a significant challenge, particularly in critical applications such as credit risk assessment. Model interpretability has spurred the development of explainable machine learning techniques that offer the promise of unveiling the rationale behind model predictions, enabling stakeholders to understand, validate, and trust the decision-making process (Salter, 2023).

1.2 Problem Statement

In an ideal digital lending ecosystem, financial institutions seamlessly apply advanced machine learning techniques to assess credit risk accurately and fairly. In this scenario, machine learning models are not only reliable but also transparent and interpretable, ensuring that lenders, borrowers, and regulatory bodies understand the factors driving lending decisions. This will facilitate efficient credit risk segmentation, resulting in customized lending solutions that maximize financial inclusion and minimize default rates.

The emergence of credit risk scoring models through machine learning has opened up avenues for more accurate credit risk evaluation. Nevertheless, the complexity of these models can obscure the factors driving the decision-making process. The challenge arises from the need to reconcile advanced machine learning tools with the demand for transparent and easily interpretable insights into consumer segmentation and risk evaluation. Additionally, the lack of a comprehensive comparative analysis of machine learning tech-

niques in the context of credit risk segmentation exacerbates the challenge. Financial institutions often struggle to select the most suitable models, leading to sub-optimal lending decisions, higher default rates, and unequal access to credit.

Bridging this gap is vital not just for lending institutions to make well-informed credit choices but also for consumers to comprehend and trust the appraisal of their credit-worthiness. Addressing these multifaceted challenges demands the development of an innovative framework that skillfully harnesses the potential of transparent machine learning tools to enhance the accuracy of credit risk scores and interpretable insights into customer segmentation. This study aims to design such a framework, thus playing a pivotal role in advancing credit risk scoring methodologies.

1.3 Research Objectives

- i To develop various machine learning tools for customer risk segmentation within digital lending.
- ii To compare accuracy between the top-performing machine learning models and the conventional statistical methods currently in use. This comparative evaluation will offer valuable insights into the potential improvements achievable through AI-driven methodologies.
- iii To adopt interpretable machine learning tools to gain an understanding of feature importance.
- iv To deploy the model for use in customer risk segmentation in digital lending.

1.4 Research questions

- i How can machine learning tools be effectively applied to enhance the precision and granularity of customer risk segmentation in digital lending?
- ii What are the specific advantages and limitations of machine learning methodologies in credit risk assessment, as revealed through a comprehensive comparative analysis of the current statistical methods in use?
- iii How can interpretability techniques contribute to a deeper understanding of credit risk assessment outcomes?
- iv How can the insights generated through these techniques provide actionable information for decision-makers in the digital lending domain, and how can this knowledge inform digital lending practices?

1.5 Research Scope

This research focuses on the implementation of machine learning techniques for customer segmentation in digital lending. It seeks to improve the precision and accuracy of customer segmentation and to explore how machine learning models can enhance credit risk assessment. The study will also delve into the practical application of explainable machine learning methods, such as SHAP values and feature importance analysis, to shed light on the factors influencing credit risk decisions. Ultimately, the research aims to contribute to the development of more transparent, data-driven, and efficient practices in credit risk assessment within the digital lending industry.

1.6 Research Limitations

One limitation of this study is the complexity of machine learning models. While they offer improved accuracy, their complexity can make them less interpretable, which may hinder the adoption of these models in real-world lending institutions.

Balancing the need for model interpretability with the desire for high predictive performance can be challenging. In some cases, achieving maximum interpretability may come at the cost of predictive accuracy and vice versa.

External factors beyond the scope of this research, such as regulatory changes and organizational constraints, may impact the practical implementation of machine learning-based customer segmentation in the digital lending industry.

1.7 Research Justification

Accurate customer segmentation is at the core of this research due to its paramount significance in the digital lending industry. As the digital lending industry continues to grow, the ability to differentiate between various borrower profiles becomes increasingly critical. The study seeks to address the pressing need for improved segmentation methods that enable lenders to offer tailored financial products, thus enhancing the overall customer experience. Furthermore, accurate segmentation plays a pivotal role in risk assessment, reducing default rates. This study aims to contribute to the evolution of the digital lending landscape and drive the adoption of more efficient, transparent, and data-informed lending ecosystem.

Chapter 2

Literature Review

2.1 Overview

Credit risk scoring plays a vital role in assessing the creditworthiness of individuals and credit risk scoring techniques have undergone significant evolution over time, driven by advancements in technology, access to data, and the need for more accurate risk assessment. This chapter will provide an overview of these advancements in digital lending, customer segmentation, machine learning tools, and model interpretability.

2.2 Digital Lending

There has been a proliferation of digital lending platforms, peer-to-peer lending networks, and mobile loan applications, providing consumers with swift access to credit while minimizing documentation burdens. This form of lending is commonly facilitated through a digital platform or mobile application, enabling borrowers to seamlessly execute the application process, access funds, and manage loan repayments entirely via the internet (*Digital Lending Market Size, Trends , Growth — Forecast - 2032* 2023).

The adoption of machine learning technologies can significantly transform and improve the digital lending landscape which creates more growth for lenders and an increase in financial inclusion to access credit amongst borrowers. These tools can analyze large

volumes of data from a borrower's credit history, mobile phone activity, and other sources to create a more comprehensive and accurate picture of their creditworthiness (*Digital Lending Market Size, Trends , Growth — Forecast - 2032* 2023).

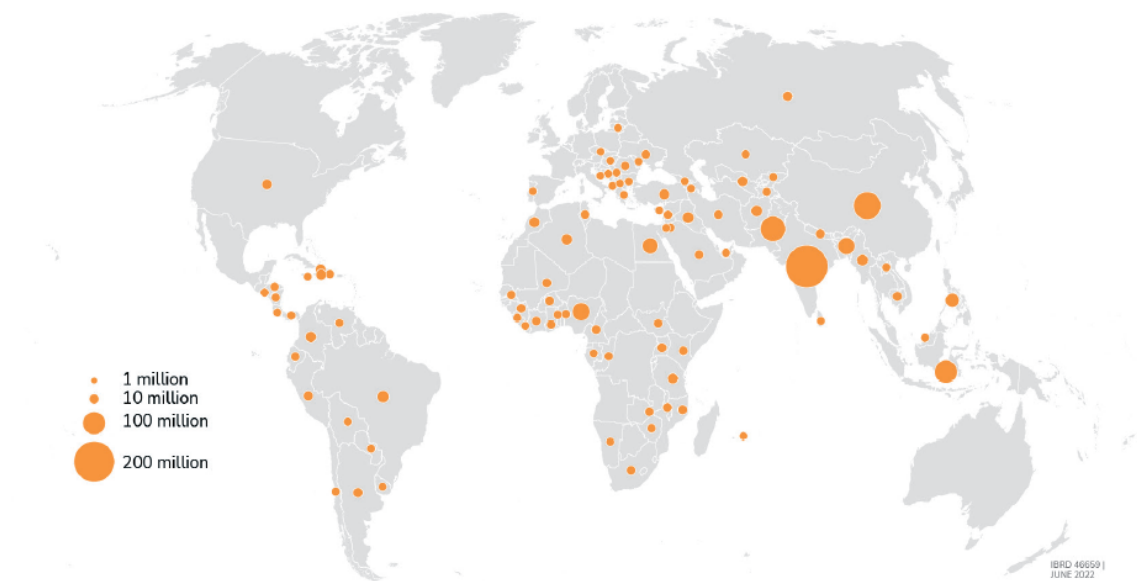
Numerous households in emerging economies do not possess established financial track records, creating challenges for companies seeking to provide credit and for individuals seeking access to credit facilities. Nevertheless, a significant portion of these households own mobile phones, generating valuable data on their financial behaviors (Björkegren and Grissen, 2019). Unbanked individuals consistently point to several key factors, including insufficient funds, geographical distance to the nearest financial institution, and a lack of necessary documentation, as their primary reasons for not holding a bank account.

Digital lending platforms and applications have successfully entered into these regions that traditional banking institutions might not adequately cater to, thereby expanding their financial reach and accessibility (Allen, Gu, and Jagtiani, 2020). Nonetheless, all these applications encounter a core challenge: *how can a lender determine the likelihood of repayment from a prospective borrower?* Traditional credit scores are a rarity among residents in developing countries. Furthermore, a significant portion of the population has limited or no interaction with formal financial institutions, with approximately 1.4 billion unbanked adults(*The Global Findex Database* 2021).

MAP 1.2.1

Globally, 1.4 billion adults are unbanked

Adults with no account, 2021



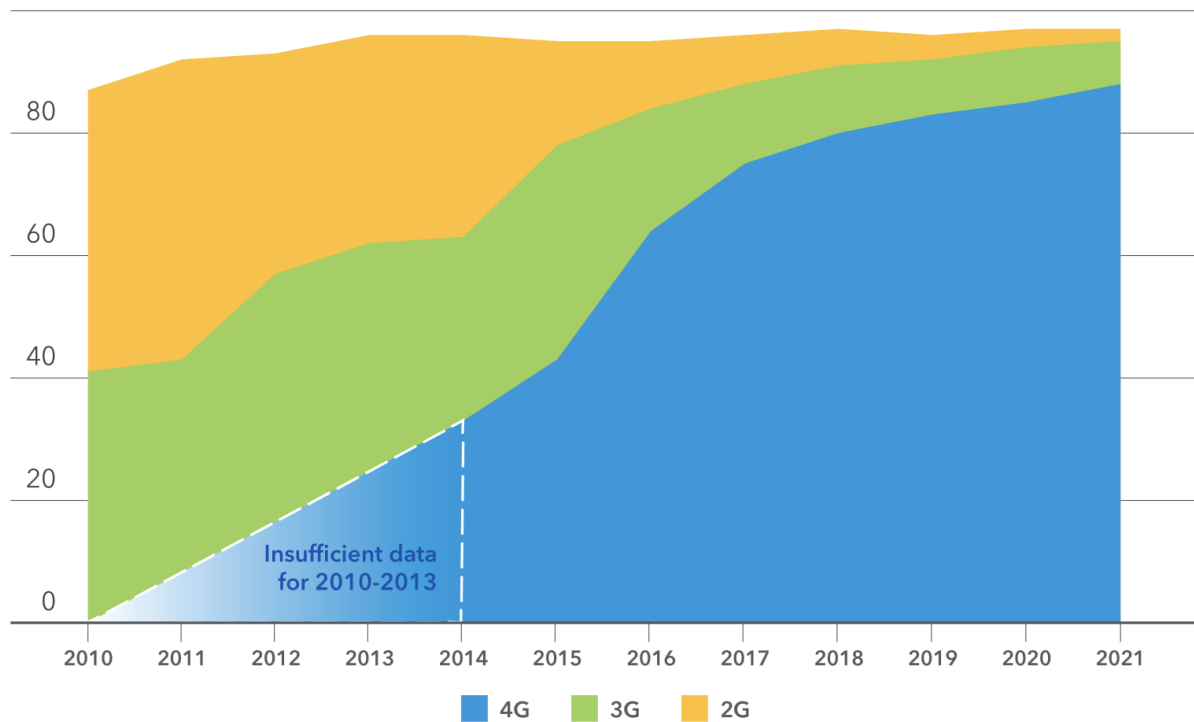
Source: Global Findex Database 2021.

Note: Data are not displayed for economies in which the share of adults without an account is 5 percent or less.

Figure 2.1: Global Connectivity Report Showing Map of Unbanked Individuals (Michael et al., 2022)

Digital lending appeals to the unbanked as these borrowers do not need to provide a handful of documentation and proof of credit history to gain access to credit. Globally, the growth of mobile phone usage has tremendously grown as seen in the figure below that shows this annual growth (Michael et al., 2022).

Percentage of the population covered by a mobile network, by generation of mobile network



Global Connectivity Report on Mobile Phone Usage (Michael et al., 2022)

By harnessing the wealth of information available through mobile money data, digital lending platforms can offer loans directly to consumers with minimal transaction costs, effectively revolutionizing the lending landscape. For instance, in Kenya, services like M-Shwari, a digital lending product launched in 2012, have pioneered instant credit access, with interest rates charged at 9% (M-Shwari 2023). This transformative approach facilitates financial inclusion and reshapes the way credit is accessed and utilized.

Björkegren and Grissen, 2019 collected data from a Telecommunications firm in a middle-income South American country that is transitioning subscribers from prepaid to post-paid plans where only 34% of adults have opened bank accounts but 89% of the households have mobile phones. The objective was to forecast the probability of repayment by utilizing behavioral attributes extracted from mobile phone usage data. Random forest and logistic regression were the selected tools for machine learning which performed better than the country's credit bureau models. The findings indicate that insights extracted from mobile phone usage can mitigate information imbalances, paving

the way for innovative, cost-effective lending approaches through machine learning tools (Björkegren and Grissen, 2019).

Using a sample of more than a million personal loans that were originated through a digital lending platform called LendingClub, a research paper by Jagtiani and Lemieux, 2019 evaluated the default drivers of Fintech loans. The likelihood of default was found to be strongly correlated with a wide range of contractual loan features, borrower characteristics, and macroeconomic variables, such as loan maturity, homeownership, loan purposes, occupation, etc. This in-depth investigation made use of LASSO, a supervised machine learning method, to identify the key variables among a huge number of probable loan default determinants (Jagtiani and Lemieux, 2019).

2.3 Machine Learning Applications in Customer Segmentation

Customer segmentation is a method for grouping consumers based on similar patterns into clusters, making it easier to manage a big client base. One of the most prevalent approaches to consumer segmentation is based on the RFM (Recency, Frequency, and Monetary) method, which helps to establish clusters based on their behavior using various clustering algorithms.

The RFM model has found extensive application in various scholarly research endeavors. Numerous researchers have employed this model for tasks such as customer segmentation, as evidenced by studies conducted by Ching-Hsue and You-Shyang, 2009 who's aim was to develop an algorithm capable of extracting all RFM patterns from customer purchase data. The results were used to derive valuable insights into customer purchasing behavior thus aiding managerial decision-making and demonstrating the advantages of using RFM analysis for examining customer purchase data within the retail sector (Ching-Hsue and You-Shyang, 2009).

In another study done by Ansari, Riasi, et al., 2016, the objective was to identify primary clusters among bank customers to facilitate customer classification and formulate more effective customer strategies. The results revealed that distinct groups of bank customers can be characterized by their loan amounts, default risk, account balances, levels of loyalty, and their profitability for the bank (Ansari, Riasi, et al., 2016).

Recency refers to how recently a consumer purchased a service or product, whereas frequency refers to how frequently a customer purchases a product, and monetary refers to how much a customer wishes to pay to purchase a product. The Total Credit score = Recency credit + Frequency + Monetary + credit on payment method + credit on return on valid/invalid reasons from a customer dataset of a company from the communication industry. The best customers were chosen with the highest credit scores (Maryani and Riana, 2017).

The research study from Maryani and Riana, 2017 also used the traditional K-Means algorithm based on the triangle inequality theorem, to improve the efficiency of the K-means clustering algorithm. The K-Means algorithm, categorized as a distance-based clustering method, partitions data into multiple groups based on numerical attributes, allowing for the exploration of inherent patterns within the dataset. Four clusters were created from the K-means algorithm: excellent customers, decent customers, average customers, and worst customers, and actions taken according to the cluster a specific consumer belongs to. (Maryani and Riana, 2017).

A research study whose focus was to segment the customers from the transaction data of 82,648 records based on the RFM model, and furthermore clustering analysis was performed by using K-Means resulted in 102 customers. 63 customers were in Cluster 1, and 39 customers were in Cluster 2. Cluster 1 had a higher average of RFM value than Cluster 2. It is anticipated that the business will be able to choose the best marketing plan by understanding the classifications of each customer (Maryani, Riana, et al., 2018).

In a research study on *CUSTOMER SEGMENTATION BY USING RFM MODEL AND*

CLUSTERING METHODS: A CASE STUDY IN RETAIL INDUSTRY, a dataset from customer loyalty cards accounts was used that had 715,328 records. A final RFM score was been obtained for each customer, for instance, an RFM score of 323 means that the customer's recency value is 3, the frequency value is 2, and the monetary value is 3 with 1 being the lowest value attained and 3 as the highest values to be attained. The K-Means clustering was also used with 4 clusters: Loyal, Regular, Advanced, and Star (Dogan, Aycin, and Bulut, 2018). The research came to the conclusion that greater customer segmentation is essential for retail businesses because it groups customers who share similar needs, wants, and habits, allowing businesses to better understand their target markets.

Unsupervised learning, a branch in Machine learning, holds a multitude of significant applications, one of which is customer segmentation with an aim to model the underlying structure or distribution in the data to learn more about the data. This involves the process of identifying distinct customer segments within the potential user base through the utilization of clustering techniques like K-means, Agglomerative, and Mean Shift. These techniques group customers together based on shared attributes such as gender, age, as well as common interests and spending behaviors (Raj et al., 2023).

In a research study on *Customer segmentation using Credit Card Data*, an unlabeled dataset was used to identify similarities using the K-Means algorithm, Density-Based Clustering, and Hierarchical Clustering. Hierarchical clustering yields a collection of clusters, where data points within the same cluster exhibit similarities. Density-based clustering relies on the fundamental notion that a cluster in data space forms a continuous area with a high density of data points, separated from other clusters by areas with notably lower data point density. Moreover, it offers a visual interpretation through a dendrogram. The K-Means algorithm had the best Silhouette Score amongst the 3 algorithms based above resulting in 4 clusters where 2 clusters were of utmost importance (Raj et al., 2023).

2.4 Machine Learning Applications in Credit Scoring

Credit scores, including FICO scores developed by Fair Isaac Corporation in the 1950s, have traditionally played a central role in credit-related determinations, particularly in credit card applications. The FICO credit score fluctuates with a person's changes in how they manage their credit accounts and bills. The primary determinants of a person's FICO score are the length of their credit history, the age of their loan accounts, and the length of time since any action on those accounts (Hindistan et al., 2019).

FICO scores have been the cornerstone of assessing an individual's creditworthiness and eligibility for various financial products and services. Depending on the scoring model, FICO scores can range from 300 to 850 or 250 to 900, and is based on 35% payment history, 30% debt burden, 15% length of credit history, 10% type of credit used and 10% of recent credit inquiries. The higher the credit score, the more creditworthy is the borrower (Allen, Gu, and Jagtiani, 2020).

In recent times, prominent Fintech players such as Alibaba and Tencent's WeChat Pay Points Credit in China have spearheaded the development of innovative credit scoring systems devised around alternative data from social media, e-commerce, mobile payment apps, mobile phone records, and more. This approach offers a holistic perspective on consumers' financial behaviors and lifestyles with the objective of bridging the credit accessibility gap for individuals who face challenges in securing loans due to their limited or nonexistent credit histories (Allen, Gu, and Jagtiani, 2020).

The primary objective of every credit scoring procedure is to summarise all accessible information about a borrower into a single score. If a borrower's score surpasses a predefined threshold, they are approved for credit; otherwise, their application is declined. Credit scoring serves a dual purpose: it is employed for evaluating new applicants, known as application scoring, and for monitoring the creditworthiness of existing borrowers to detect alterations in their financial behavior, referred to as behavioral scoring (Chitambira, Engstrom, and Ster, 2022).

2.4.1 Credit Scoring Models categories

Credit scoring models can be broadly categorized into three main domains: statistical, machine learning, and deep learning.

Statistical Tools

Historically, the methods that were most frequently employed to create credit scoring models were linear discriminant analysis (LDA) and logistic regression. Both benefit from being widely available in statistical software packages and are conceptually simple. The coefficients and numerical scores of the qualities were combined to provide single contributions, which were then aggregated to get an overall score (Crowther and Bolton, 2010). The disadvantage of LDA is that it requires normally distributed data, but in reality, the credit data are often not normally distributed.

Logistic regression is adept at quantifying the likelihood of credit default or approval based on a set of pertinent factors. It serves as a cornerstone in predictive modeling, offering valuable insights into binary classification tasks and aiding in sound decision-making processes (Kleinbaum and Klein, 2010).

A study from a banking dataset with 83,685 records and 48 features to obtain the most predictive variables that will be used in logistic regression revealed that 7 variables were most useful with high predictive power. The study concluded that the creation of credit scoring models involves more than merely statistical techniques and the numbers should be carefully evaluated to verify that they make sense](Crowther and Bolton, 2010).

A study on the *Performance Analysis of Credit Scoring Models on Lending Club Data* used a peer-to-peer lending platform dataset to conduct an in-depth model performance comparative analysis of classification tools. The dataset had 227,963 observations and 115 variables though only 23 variables were used following data pre-processing steps. Logistic regression significantly outperformed Random Forest and it ranked the best classifier

in comparison to the Linear Discriminant Analysis. The study concluded that Logistic regression and Linear discriminant analysis are proper classification algorithms for credit scoring since Logistic regression, Artificial neural network, and Linear discriminant analysis are the three best algorithms based on the Lending Club data. Random forest ranked the fifth best classifier in this study (Polena, 2017)

Additionally, a study on *Alternative Credit Scoring and Classification Employing Machine Learning Techniques on a Big Data Platform* on Lending Club data. The study revealed that in comparison to Decision Tree and Random Forest, Logistic Regression offers the best accuracy, precision, and recall scores. Taking into account the widespread perception that the most widely used and recognized methods for credit assessment are Logistic Regression and Random Forest. Not only does Logistic Regression perform better, but it also uses minimal resources in generating an output (Hindistan et al., 2019).

Statistical techniques continue to be widely employed today offering lenders the advantage of utilizing principles such as sample estimators, confidence intervals, and statistical inference. This framework enables scorecard developers to assess the effectiveness of models in distinguishing between borrowers and also to identify which borrower attributes play a more significant role in explaining borrower behavior (Chitambira, Engstrom, and Ster, 2022). They rely on mathematical models and statistical principles to establish relationships between random and non-random variables (Salter, 2023).

Machine Learning Tools

As per the recent *European Discussion Paper on Machine Learning for IRB (Internal Ratings-Based) Models* to calculate regulatory capital for credit risk, Machine Learning (ML) models used in the Basel II regulation's IRB approach can be effective in enhancing risk distinction by enhancing the discriminative capacity of the model and offering valuable tools for recognizing all pertinent risk factors, including potential relationships among them. ML is also useful in improving data collecting and preparation processes, such as cleansing input data or offering a tool for data treatment and data quality

checks, and could also be used to detect outliers and repair errors. Additionally, ML models are used in providing strong systems for model validation and monitoring and also perform stress testing such that the predictions from an ML model are not affected when the environment changes (*EBA DISCUSSION PAPER ON MACHINE LEARNING FOR IRB MODELS* 2021).

The decision tree machine learning algorithm uses a hierarchical data structure that employs the divide-and-conquer strategy. When making predictions for a specific instance, we categorize an observation within a particular region based on the class that appears most frequently among the training observations in that same region. A dataset from customer default payments in Taiwan in October 2005 was used to adopt machine learning algorithms in credit scoring. A Grid Search was conducted to identify the parameter values that would prevent the tree from over-fitting. The study revealed that the decision performed the fourth best in comparison to other ML algorithms (Chitambira, Engstrom, and Ster, 2022).

Random Forest represents a significant variation of bagging (bootstrap aggregating), constructing a substantial ensemble of uncorrelated trees and subsequently averaging them. Random forests offer the advantage of being easier to train and fine-tune. In the construction of a random forest, multiple decision trees are created using bootstrapped training samples, where bootstrapping involves re-sampling through random selection with replacement. In the study mentioned above on the Taiwan dataset, Random Forest emerged as the best-performing model based on accuracy and this finding is consistent with other results from previous research (Chitambira, Engstrom, and Ster, 2022)

In the context of credit scoring, the K-nearest neighbors approach employs a specific metric to assess the distance between two applicants within the application data space. By utilizing a sample of historical applicants as a benchmark, a new applicant is categorized as either favorable or unfavorable based on the distribution of "good" and "bad" applicants among the k-nearest neighbors from the representative sample. In essence, the classification of the new applicant relies on the characteristics of their closest neighbors in the sample (Chitambira, Engstrom, and Ster, 2022).

eXtreme Gradient Boosting (XGBoost) stands out as a tree-boosting algorithm initially crafted by Tianqi and Carlos, 2016. What sets XGBoost apart from conventional gradient-boosting models is its heightened focus on regularization techniques, yielding more generalized models that are less susceptible to over-fitting and exhibit enhanced performance characteristics. In a study on *Explainable Artificial Intelligence and its Applications in Behavioural Credit Scoring*, the XGBoost performed strongly and it identified payment behavior features as the most important. Thus, incorporating more features that capture a customer's account behavior may be the greatest way to improve default prediction (Salter, 2023).

Gradient Boosting Machine (GBM) stands out in the realm of supervised learning when it comes to structured datasets. Boosting follows a sequential model-building approach that commences with a "weak learner," a model slightly superior to random guessing, and progressively refines its predictions by rectifying the errors of the preceding model at each iteration. In a study titled *Beyond Accuracy in Artificial Intelligence Based Credit Scoring Systems*, GBM exhibited the least impressive performance which notably lagged behind the top-performing Deep Learning (DL) model. The study preferred GBM as the preferred model of choice over DL due to its ease of usage and offers strong prediction accuracy (Schmitt and Cummins, 2023).

Machine learning leverages empirical data to uncover patterns and relationships and with the ongoing discussions, the future of ML models will become more widespread (Salter, 2023).

Deep Learning

Deep learning (DL) also known as hierarchical learning has emerged as one of the transformative machine learning models in use. Its rise can be attributed to three pivotal factors: advances in data availability, processing power, and optimization algorithms (Schmitt and Cummins, 2023).

In an empirical study conducted on *Ensemble Learning or Deep Learning? Application to Default Risk Analysis*, it concluded that DL tends to yield superior performance when utilizing a single-layer neural network on average in the domain of credit scoring and structured data. Consequently, if one wishes to employ a neural network, selecting a single-layer configuration is the most prudent decision. This choice not only substantially reduces computational complexity, enhancing sustainability, but also enhances accuracy (Schmitt and Cummins, 2023).

Artificial Neural Networks (ANNs) draw inspiration from how the human brain cells function. Much like the human brain, ANNs exhibit the capacity to grasp complex patterns through exposure to diverse examples. The learning process of a neural network encompasses several steps: calculating the output, fine-tuning the weights based on the disparity between the outputs and desired targets, and then iteratively repeating this procedure (Chitambira, Engstrom, and Ster, 2022). Through multiple iterations of this learning process, the network gradually converges on the optimal weight values. However in a study by Chitambira, Engstrom, and Ster, 2022, the ANN scored the second least in performance/.

Recurrent neural networks (RNNs) represent a specialized variant of multi-layer perceptrons that find their strength in classification tasks. A perceptron is defined as the basic processing element that has its inputs coming from the environment or from the output of other perceptrons. From the dataset on customer (Chitambira, Engstrom, and Ster, 2022).

Conventional RNNs suffer from a well-documented limitation known as the vanishing/exploding gradient problem, especially as the temporal lag between inputs increases. To address this challenge, Long Short-Term Memory (LSTM) neural networks were introduced as a specialized variant of RNNs designed to mitigate the gradient problem. In a study by Salter, 2023, the LSTM outperformed the LR model, though not statistically significant. The problem is that model performance is not the only factor considered in credit scoring. Regulators and users want explainable decisions, and here is where the LSTM model falls short (Salter, 2023).

2.5 Machine learning explainability

Explainable Artificial Intelligence (XAI) constitutes a subfield within computer science dedicated to rendering the actions of artificial intelligence algorithms comprehensible to human users. Many machine learning and deep learning models have garnered the label of "black boxes," signifying that the intricacies governing the transition from input to output remain mysterious to humans. With the progression of models towards heightened complexity and improved predictive capabilities, the human capacity to fathom the decision-making mechanisms progressively diminishes (Salter, 2023).

2.5.1 SHAP values

In the realm of machine learning, SHAP values was introduced for the interpretation of predictions. This innovation was prompted by the challenge of reconciling accuracy and interpretability in ensemble and deep learning models, which were traditionally viewed as a trade-off. The SHAP method initiates its journey from the baseline of expected model predictions, devoid of any prior knowledge about the features. Through a meticulous iterative procedure, it assigns attributions to features based on the alterations they induce in the expected model prediction when conditioned on a particular feature (Ariza-Garzón et al., 2020).

Combining SHAP with nonlinear models allows for the interpretation of explanations pertaining to intricate, nonlinear relationships (Ariza-Garzón et al., 2020). In a study by Salter, 2023, it concluded that integrating XGBoost with the post-hoc SHAP technique yielded outcomes that elucidated the significance of individual variables in the prediction process. Consequently, this approach offers a practical means of enhancing the interpretability of otherwise opaque models and identifying the key contributing factors. However, when attempting to clarify LSTM predictions, it was unsuccessful, as SHAP failed to generate satisfactory explanations for the model's forecasts (Salter, 2023)

2.5.2 Permutation Feature Importance

This method provides a comprehensive global explanation for interpreting a model by assessing the impact of altering (permuting) the values of a feature. The significance of a feature is established by evaluating whether modifying its values leads to an increase in model error. If such a change indeed results in increased error, it signifies the model's dependency on the feature's values for accurate predictions. Conversely, if there is no discernible shift in error, it implies that the feature's values made no meaningful contribution to the model and are, therefore, deemed unimportant Molnar, 2020.

A case study on *Predictive case-based feature importance and interaction* concluded that the use of permutation feature importance offered an intricate examination of the functions of features and the interplay among these features within a predictive model (Sejong, 2022).

2.5.3 Partial Dependence Plots

Partial Dependence Plots (PDPs) serve as a valuable tool in assessing feature importance within machine learning algorithms as they provide insights into how each feature contributes to the decision-making process, assuming independence among features. One of the notable advantages of PDPs is their ability to reveal the impact of single-feature variations on predictions and its computational efficiency. However, it's essential to acknowledge that PDPs may present challenges for non-experts due to the intricate visualizations they produce (Hashemi, 2023).

In a credit risk scoring and segmentation scenario, PDPs can be applied to examine the influence of various applicant attributes, such as education level, on creditworthiness. For instance, by keeping all other features constant and varying the education level across categories like high school, B.Sc., M.Sc., and Ph.D., one can assess how the changes in the level of education can impact credit risk predictions.

2.6 Summary Findings

Research Study	Author	ML Tool and Evaluation	Model Explainability
Explainability of a Machine Learning Granting Scoring Model in Peer-to-Peer Lending	(Ariza-Garzón et al., 2020)	ML Tools: Logistic Regression, Decision Tree, Random Forest and XGBoost. Evaluation Tools: Accuracy, AUC and Kolmogorov-Smirnov statistic (KS)	SHAP
Credit Scoring using Machine Learning Approaches	(Chitambira, Engstrom, and Ster, 2022)	ML Tools: Logistic Regression, Decision Tree, Support Vector Machine, Artificial Neural Network, Random Forest, and K-Nearest neighbor. Evaluation metrics: Accuracy, Precision, Recall, F1, AUROC and Brier Skill Score	
Explainable Artificial Intelligence and its Applications in Behavioural Credit Scoring	(Salter, 2023)	ML Tools: Logistic Regression, XGBoost, and LSTM. Evaluation Tools: ROC-AUC, Accuracy, Precision, F1, Brier Score, Gmean and McNemar Test	SHAP and Permutation importance
Customer Segmentation Using Credit Card Data Analysis	(Raj et al., 2023)	ML tools: K Means-Clustering, Density-Based Clustering, Hierarchical Clustering. Evaluation tools: Silhouette score	
Beyond Accuracy in Artificial Intelligence Based Credit Scoring Systems: Explainability and Sustainability in Decision Support	(Schmitt and Cummins, 2023)	ML Tools: Deep learning (DL) and Gradient Boosting (GB). Evaluation Tools: AUC, Accuracy, F-Score, and Logloss	Feature Importance Plots, SHAP and Partial Dependence Plots

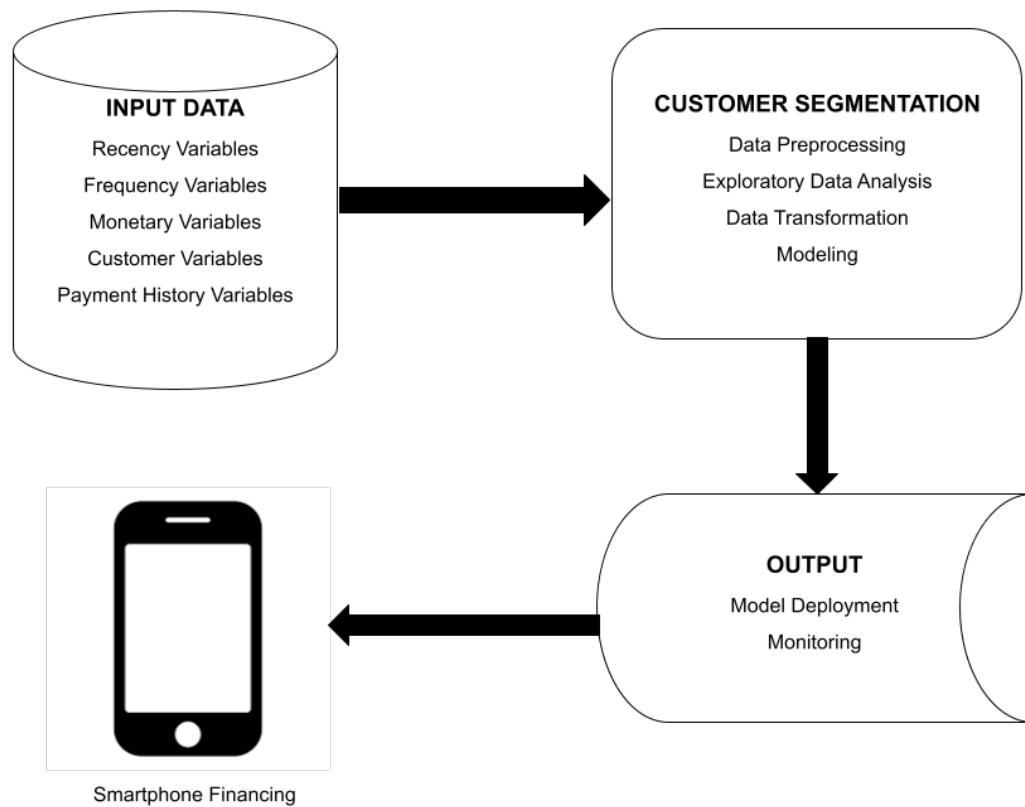
Table 2.1: Influential papers

<https://www.overleaf.com/project/650f21b21352b16f41f5a1ab>

2.7 Research gap

This research identifies a critical gap in the application of unsupervised machine learning techniques to telecommunication data for customer risk segmentation in digital lending. While machine learning has shown promise in credit scoring, it predominantly focuses on supervised learning with traditional financial data. This study seeks to harness the potential of alternative data sources and evaluate the interpretability and practicality of machine learning models in the digital lending sector. By addressing these gaps, it aims to enhance customer segmentation accuracy and contribute to the field's advancement.

2.8 Conceptual Framework



Conceptual Framework

Chapter 3

Methodology

3.1 Research design

A research design serves as a well-structured blueprint for attaining the objectives defined in a study. For this study, the CRISP-DM (CRoss Industry Standard Procedure in Data Mining) framework will be adopted to provide a structured format for managing the data science project.

The first step is identifying the measurable objectives that will be used as guidelines throughout the study. This understanding involves engaging with the stakeholders to determine the key business goals and identify the current gaps and challenges. Further research is important in reviewing past relevant work through literature reviews.

The subsequent stage involves gaining a deeper understanding of the data. This dataset comprises historical records obtained from a telecommunications company. The structure and quality of the data will be analyzed to confirm that the data meets the objectives of the study. Any identified data gaps or issues will be addressed. Most importantly, the regulatory and compliance aspects related to the data will be keenly observed.

After the data understanding phase, the next step is data preparation, which encompasses data cleaning, transformation, and feature engineering to ensure the dataset is of high quality for machine learning modeling. During this stage, we will identify anomalies,

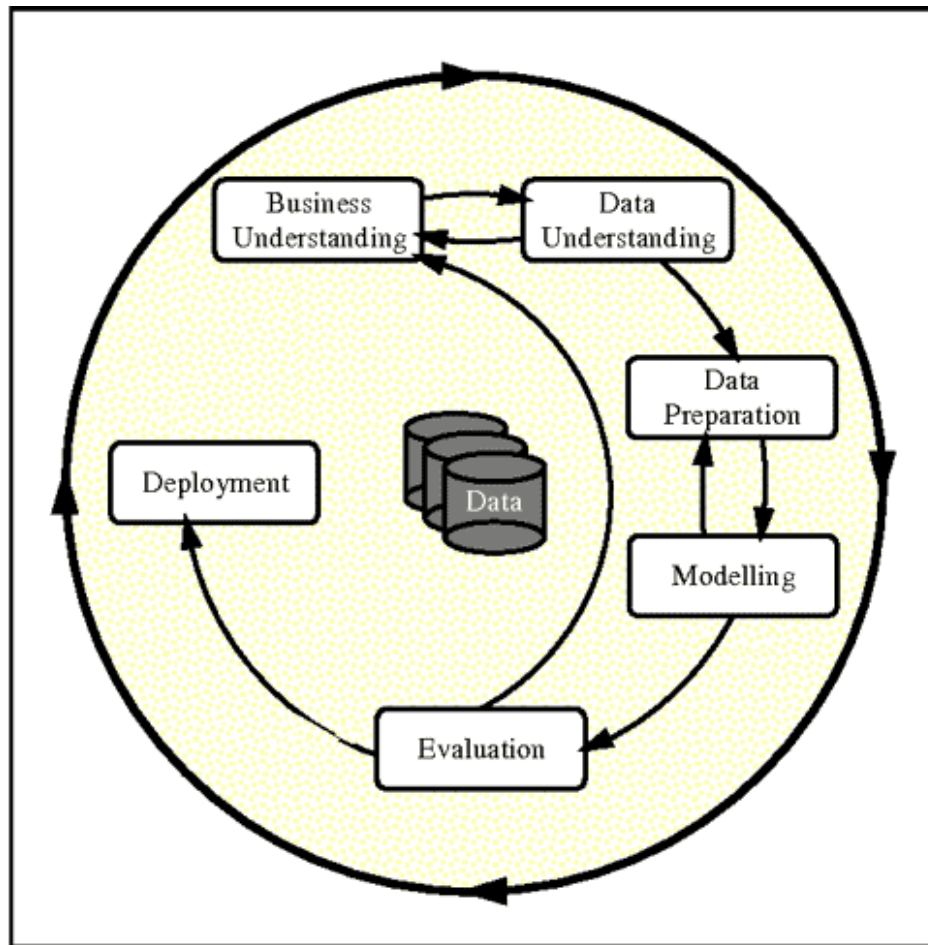


Figure 3.1: CRISP DM Framework Rüdiger and Jochen, 2000

such as missing values and outliers, and perform a thorough analysis to determine their root causes.

The modeling phase follows and it focuses on building and training machine learning models. It encompasses a systematic exploration of diverse machine-learning algorithms and methodologies to identify the optimal machine-learning model. Cross-validation techniques will be adopted to fine-tune hyper-parameters and prevent overfitting.

The performance of the selected models will then be evaluated using accuracy metrics such as accuracy, precision, recall, and F1-score which are useful in determining the best-performing models and assessing their suitability for making predictions on new, unseen data. Considering we are including a comparative analysis of the current scoring model being used, the credit scoring results from the best-performing model will be

analyzed with the current scoring model. This will determine which of the two models should be adopted for future credit risk scoring segmentation.

Given the project's focus, a significant effort will be dedicated to model interpretability using techniques like SHAP (SHapley Additive exPlanations), LIME (Local Interpretable Model-Agnostic Explanations), and partial dependence plots to provide transparent explanations of credit risk assessments.

3.2 Data sources and Description

The research data will be obtained from a digital lending platform with access to customer telecommunications data. The dataset encompasses various attributes, including customer demographics (such as age, gender, employment status, and income) and mobile phone usage patterns, which encompass data on the number of incoming and outgoing calls, expenditures on telephone calls, usage of internet data bundles, and other related services. Furthermore, the dataset includes information about transactions made with other merchants and the social connections of customers, quantifying both the value and quantity of these interactions conducted through mobile money payment platforms. Additionally, we will explore and incorporate any other relevant alternative data sources applicable within the context of digital lending.

3.3 Data Preprocessing

This step involves cleaning data, exploration and transformation to ensure the data is of high quality for modeling.

Missing values will be identified and various approaches will be used to deal with them. One of the approaches to be considered is Mean/Median Imputation that involves replacing missing values with the mean or median of the feature. Another approach is For-

ward/Backward Fill that's applicable for time series data that uses values from adjacent time points to fill the missing values. Lastly, Interpolation where linear or polynomial interpolation is used to estimate missing values. In scenarios where columns have more than 30% missing values, these columns will be removed from the dataset.

Outliers are extreme values and these greatly affect the model performance. These will be identified using Z-score that identified data points that are significantly different from the mean or using visuals like box-plots and histograms. Depending on the column, the outliers will be deleted or transformed using log transformation to reduce the impact on the model performance.

Duplicated values will be identified and deleted from the dataset.

3.4 Exploratory Data Analysis

Exploratory Data Analysis (EDA) is an essential initial phase in the data preparation process, providing invaluable insights into the dataset's characteristics and quality. This will involve examining and understanding the dataset before performing any modeling.

Descriptive statistics offer a comprehensive overview of the dataset, summarizing key numerical attributes and highlighting central tendencies, spread, and distribution characteristics. These include mean that provides the average value of numerical features and offers insight into the dataset's overall trend. Median which is a measure of the central position in the data, indicating the middle value when data is sorted and is robust to outliers. Standard deviation that quantifies the degree of data dispersion around the mean. A high standard deviation suggests a wider spread of data values and finally the range is the difference between the maximum and minimum values, reflecting the data's overall variation. Categorical features will be explored by counting the unique categories within each feature. Understanding the diversity of categories is essential for categorical data preprocessing.

Visualizing data distributions is a powerful method to uncover patterns, anomalies, or outliers within the dataset. This analysis involves creating various graphical representations to gain insight into the distribution of numerical features, such as: Histograms that provide a graphical representation of the data's frequency distribution, allowing for the identification of data clusters or irregularities. Density Plots that visualize the probability density function of numerical features, offering a smoothed view of data distributions. They are particularly useful for identifying multi-modal or skewed distributions. Lastly, Box plots provide a summary of the data distribution's key characteristics, including median, quartiles, and potential outliers. They are helpful for detecting skewness and extreme values.

Correlation analysis seeks to uncover relationships between numerical features that will be calculated to assess the degree and direction of associations among numerical attributes. Understanding these relationships is pivotal for feature selection and variable prioritization in subsequent modeling. The Pearson correlation coefficient measures linear relationships between continuous variables. It ranges from -1 (perfect negative correlation) to 1 (perfect positive correlation), with 0 indicating no correlation. Whereas, the Spearman's rank correlation assesses the strength and direction of monotonic associations between variables, making it robust to outliers and nonlinear relationships.

3.5 Data Transformation

Data transformation is a critical step in preparing the dataset for machine learning. It involves creating new features, scaling numerical features, and converting categorical data into a format suitable for machine learning models.

Feature engineering is the process of creating new features from existing data to enhance the performance and predictive power of machine learning models. In this study, feature engineering will focus on generating customer segmentation variables and relevant behavioral indicators based on domain knowledge.

Numerical features often have varying scales, which can lead to biases in machine learning algorithms. Scaling or normalization is the process of adjusting feature values to ensure they fall within a consistent range. Min-Max scaling is a normalization technique that scales features to a specific range, typically between 0 and 1. It is useful when the exact range of the data is unknown or when algorithms, such as support vector machines and k-nearest neighbors, require features to be within a certain range. Standardization (Z-Score Scaling) is also another method that transforms features to have a mean of 0 and a standard deviation of 1. Robust scaling normalization technique is robust to outliers and uses the median and interquartile range to scale features. It's a good choice when outliers are present.

3.6 Machine Learning Models

In the context of customer risk segmentation within digital lending, the selection of machine learning tools plays a pivotal role in the success of the project. the chosen machine learning models are listed below.

3.6.1 K-means Clustering

The K-Means algorithm is the most popular and simpler machine learning algorithm that uses an iterative process to try and divide the dataset into various "k" numbers of redefined and non-overlapping clusters, with each data point belonging to only one subgroup based on their shared characteristics. The number of clusters is determined through various methods such as the Elbow Method or the Silhouette Method that can help you choose an appropriate value for 'k' based on the data's characteristics (Maryani, Riana, et al., 2018).

The mean value of the objects in each cluster serves as the cluster's center (centroid), inputs: D, a data collection with n objects, and k, the number of clusters. The pro-

cedure is as follows: randomly select k objects from D to serve as the initial cluster centers; repeatedly (re)assign each object to the cluster to which it is most similar based on the mean value of the objects in the cluster; update the cluster means, that is, compute the mean value of the objects for each cluster, and repeat this process until there is no further change (Raj et al., 2023).

3.6.2 Hierarchical Clustering

Hierarchical Clustering represents an unsupervised machine learning technique used for the grouping of similar objects into clusters. This algorithm yields a collection of clusters, where data points within the same cluster exhibit similarities. Moreover, Hierarchical Clustering offers a visual interpretation through a dendrogram. There are two main variants of Hierarchical Clustering: Agglomerative Hierarchical Clustering and Divisive Hierarchical Clustering (Raj et al., 2023).

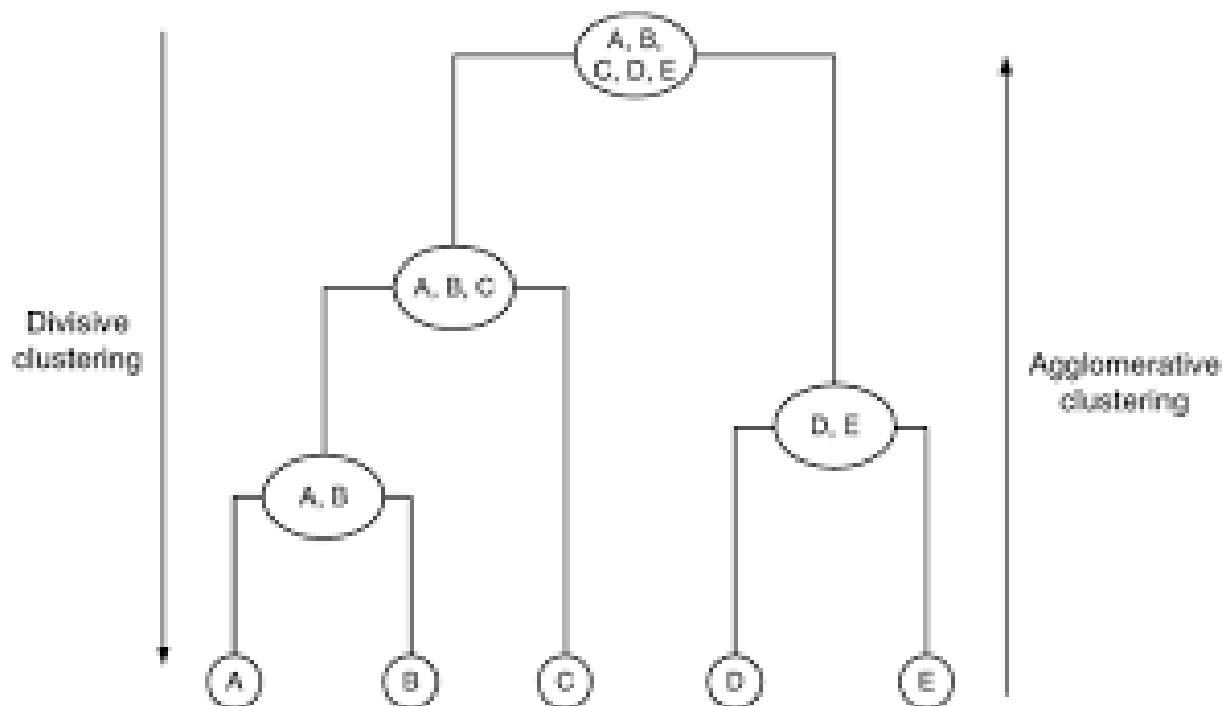


Figure 3.2: Hierarchical Clustering (Raj et al., 2023)

Agglomerative Hierarchical Clustering is a commonly used methodology that involves a bottom-up approach, where individual data points are initially treated as separate clusters and are then progressively merged into larger clusters based on their similarities.

Conversely, Divisive Hierarchical Clustering starts with all data points in a single cluster and recursively divides them into smaller clusters, ultimately creating a hierarchy of clusters. These methods provide valuable insights into the hierarchical relationships and structures within complex datasets (Raj et al., 2023).

3.6.3 Density-Based Clustering

Density-Based Clustering represents a category of unsupervised learning techniques aimed at identifying distinct clusters within a dataset. This approach relies on the fundamental notion that a cluster in data space forms a continuous area with a high density of data points, separated from other clusters by areas with notably lower data point density. One of the fundamental algorithms in density-based clustering is the Density-Based Spatial Clustering of Applications with Noise (DBSCAN). DBSCAN stands out as a foundational method for uncovering clusters of various shapes and sizes within extensive datasets, even when the data contain noise and outliers. (Raj et al., 2023):

minPts: This parameter signifies the minimum number of data points required to be in close proximity to each other for a specific region to be recognized as dense.

eps (ϵ): Epsilon, or the distance measure, plays a crucial role in determining which data points are considered neighbors of a given point based on their distance from one another.

DBSCAN's flexibility in capturing clusters of varying densities and shapes, as well as its ability to handle noisy data, makes it a valuable tool in data analysis, clustering, and pattern recognition. By adjusting the minPts and ϵ parameters appropriately, DBSCAN offers a versatile approach to uncovering meaningful clusters within complex datasets, contributing to a deeper understanding of the underlying data structure (Raj et al., 2023).

3.6.4 Logistic Regression

Logistic Regression (LR) is fundamentally rooted in linear regression, a technique that utilizes a linear combination of input features to forecast continuous values. Historically, this statistical technique has found extensive use in various domains, including credit scoring, where it is adept at quantifying the likelihood of credit default or approval based on a set of pertinent factors. It serves as a cornerstone in predictive modeling, offering valuable insights into binary classification tasks and aiding in sound decision-making processes (Mathias, Stefan, and Asil, 2020).

One of the advantages of logistic regression in supervised learning lies in its inherent explainability. Linear regression offers transparent predictions, as the model is essentially a weighted function of its input variables. The weighted coefficients are transformed into an odds ratio, effectively showcasing the influence of each variable on the overall probability (Molnar, 2020).

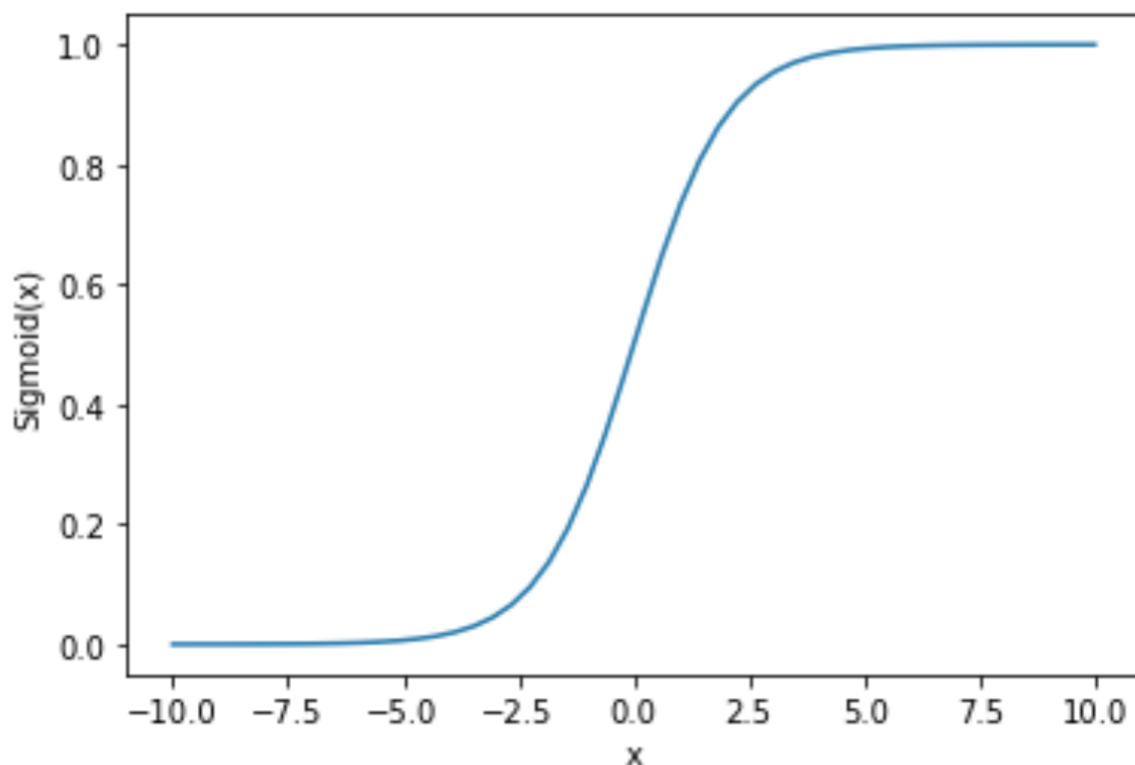


Figure 3.3: Logistic Regression Sigmoid Function (Kleinbaum and Klein, 2010)

Within the context of credit risk assessment, this sigmoidal relationship mirrors the interplay between independent variables and the dependent variable. For instance, an individual's current debt level can serve as a pertinent indicator of the likelihood of default. When the debt level is at zero, the risk of default also stands at zero. However, as the debt level progressively increases, the risk exhibits a linear growth until it reaches a critical point where the probability of default becomes virtually certain (Salter, 2023).

3.6.5 Random Forest

Random forests is a significant extension of bagging (bootstrap aggregating) involving constructing a sizable ensemble of uncorrelated decision trees and subsequently averaging their outputs. This technique is well-regarded for its simplicity in training and tuning, making it a popular choice in customer segmentation. When creating a random forest, multiple decision trees are generated from bootstrapped training samples, which are created by random sampling with replacement. During the tree-building process, a random subset of m predictors is chosen from the complete set of p predictors as split candidates. Typically, m is approximately equal to the square root of p ($m \approx \sqrt{p}$). This approach prevents the algorithm from relying on a majority of predictors for each split, thereby reducing the likelihood of correlated trees, ensuring that bagging effectively reduces variance (Chitambira, Engstrom, and Ster, 2022).

3.6.6 XGBoost

It swiftly gained recognition for delivering revolutionary results on tabular datasets at the time of its inception, and since then, it has retained its status as a go-to tool in various machine-learning domains. Notably, it has been reaffirmed through extensive research as evidenced in a paper titled *A dynamic credit scoring model based on survival gradient boosting decision tree approach* (Xia et al., 2021).

3.7 Evaluation Metrics

Accuracy serves as a statistical indicator of a classification test's proficiency in correctly identifying or excluding a condition. It represents the proportion of accurate predictions among the total cases examined. The formula for accuracy is given by:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (3.1)$$

Here, TP stands for True Positives (true non-defaults), FP for False Positives (false non-defaults), TN for True Negatives (true defaults), and FN for False Negatives (false defaults).

Precision reflects the classifier's ability not to incorrectly classify a negative sample as positive, essentially minimizing false positives. The formula for precision is as follows:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (3.2)$$

Recall measures the proportion of positive instances that were correctly identified among all instances classified as positive, aiming to reduce false negatives (false non-defaults). The recall formula is given by:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3.3)$$

The F1 score represents the harmonic mean of precision and recall and is calculated as:

$$F1 = \frac{2 \cdot \text{Recall} \cdot \text{Precision}}{\text{Recall} + \text{Precision}} \quad (3.4)$$

The AUROC (Area Under the Receiver Operating Characteristic Curve) is a graphical representation of a binary classification model's performance on the positive class.

The AUROC quantifies a classifier's ability to discriminate between defaulters and non-defaulters. A no-skill classifier has an AUROC of 0.5, while a perfect skill classifier achieves an AUROC of 1.

The silhouette score assesses the quality of clusters created through unsupervised learning techniques such as clustering. It measures how similar an object is to its own cluster (cohesion) compared to other clusters (separation). The silhouette score can range from -1 to 1, where higher values indicate better-defined clusters. In the context of customer segmentation, it helps evaluate the appropriateness of the segmentation.

3.8 Machine Learning Explainability

The study seeks to identifying the contributing variables to the model predictions. Its primary goal is to make the decision-making processes of these models understandable to humans, enabling users to understand how and why specific predictions or classifications are made.

SHapley Additive exPlanations (SHAP), named after the American mathematician Lloyd Shapley, originated from the realm of game theory and are designed to elucidate the individual contributions of actors to a collective outcome. This framework helps us to analyze the influence of each feature on the model's prediction.

Permutation Feature Importance offers a detailed explanation for model interpretation by examining how changing feature values affects the model's performance. It gauges a feature's importance by determining whether altering its values causes a change in model error.

Partial Dependence Plots (PDPs) are a useful instrument for evaluating feature significance in machine learning models as they offer insights into the contributions of each feature to the decision-making process, assuming independence among features.

3.9 Model Deployment

The model deployment will take place within a simulated lending institution environment and assessing its impact on credit risk assessment decisions. An ongoing monitoring of the model's performance to ensure its continued accuracy and the consistent production of transparent and understandable outcomes will be maintained.

The model will be continuously monitored in its performance to ensure that it continues to make accurate predictions and the outcomes produced are explainable, and consistent. Practical obstacles such as model deployment, integration into existing procedures, and potential stakeholder opposition will be documented. Recommendations for successful implementation will be presented, taking benefits and considerations into account.

The expected data product is envisioned to be a dynamic and user-friendly credit risk segmentation system. It takes input data from loan applicants or existing borrowers and, through the utilization of machine learning models, generates credit risk scores and risk segmentations in a transparent and interpretable manner. The components of the data product include an input interface that will collect the relevant information from the applicant. The machine learning model will then process the input data to predict a credit score which will be the output where the applicant will be categorized into a risk group.

Other than predicting the credit scores, the data output will also offer interpretability features that will explain the factors influencing the assigned credit risk score for each applicant, making it easier for both lenders and borrowers to understand why a particular risk assessment was made.

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